

SUGGESTED MARKING
SCHEME

JUNE 2020

CHEMISTRY 6031/3

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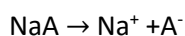
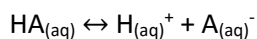
CONTACTS: 0774 957217

- 1(a)(i) order wrt: 1. RCHO - first order (by calculation/ inspection)
 2. H^+ - zero order (changing $[H^+]$ has no effect on rate)
 3. CN^- - first order

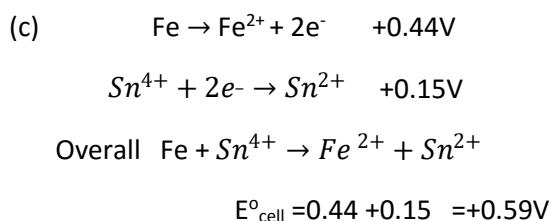
$$\begin{aligned} \text{(b)(i)} \quad H^+ &= \sqrt{Ka} \times [\text{acid}] \\ &= \sqrt{6.3 \times 10 \times 0.5} \\ &= \underline{1.254990039 \times 10^{-4}} \end{aligned}$$

$$\begin{aligned} \text{Ph} &= -\log(1.25499039 \times 10^{-4}) \\ &= 3.90 \end{aligned}$$

(ii) let benzoic acid be HA



- Maintains pH in acidic range / buffer maintained in acidic range
- Bacteria is dies in acidic conditions



Observations; - stn changes to dirty green solution

— Sn^{+4} is reduced by Fe to Sn^{2+}

2(a) Atomic radius – the halved distance between two identical nuclei.

Shielding effect-reduction of nuclear attraction to outer layers by inner layers of electrons.

(b)(i) - increase in nuclear charge across period

- yet there is no opening of new shells

(ii) – increase in nuclear charge

-number of electrons does not change

(iii) - anions gain one more shell by accepting electrons

(iv) - P^{3+} gains one more shell than P

(v) – repulsion between outer shell and inner shells of electrons in Ar

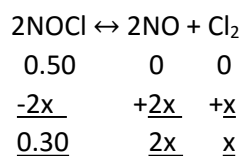
(vi) – PCl_5 has more electrons than PCl

- Therefore, PCl_5 has stronger Van der Waals forces

(vii) –P has a bigger atomic radius than N

-Hence P-H bonds are longer and therefore weaker than N-H bonds.

(c)



aside

$$0,50 - 2x = 0.30$$

$$X = 0.10$$

$$1. NO - 2(0.10) = 0.20 \text{ moles}$$

$$2. Cl_2 - 0.1 \text{ moles}$$

Partial pressures

$$\text{Total moles} = 0.3 + 0.2 + 0.1 = 0.6 \text{ moles}$$

$$P_{(NOCl)} = \frac{0.3}{0.6} \times 2.16 = 1.08 \text{ atm}$$

$$P_{(no)} = \frac{0.2}{0.6} \times 2.16 = 0.72 \text{ atm}$$

$$P_{(Cl)} = \frac{0.1}{0.6} \times 2.16 = 0.36 \text{ atm}$$

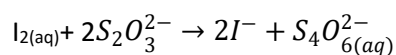
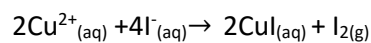
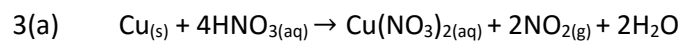
(i)

(i)

$$K_p = \frac{P(NO)^2 \times P(Cl_2)}{P(NOCl)^2}$$

$$= \frac{0.72^2 \times 0.36}{1.08^2}$$

$$= 0.16 \text{ atm}$$



Moles of I_2 liberated

$$n(\text{S}_2\text{O}_3^{2-}) = \frac{25}{1000} \times 0.1$$

$$= 0.0025 \text{ moles}$$

Molar ratio $\text{I}_2 : \text{S}_2\text{O}_3^{2-}$

$$1 : 2$$

$$n(\text{I}_2) = \frac{0.0025}{2}$$

$$= 0.00125 \text{ moles}$$

(ii) molar ratio 1 : 2

$$n(\text{Cu}^{2+}) = 0.00125 \times 2$$

$$= 0.0025 \text{ moles (in } 25\text{cm}^3)$$

$$0.0025 \rightarrow 20\text{cm}^3$$

$$n(\text{Cu}^{2+}) \rightarrow 200\text{cm}^3$$

$$n(\text{Cu}^{2+}) = \frac{200}{20} \times 0.0025$$

$$= 0.025 \text{ moles in } 200\text{cm}^3$$

(iii) mass of $\text{Cu}^{2+} = \text{Mr} \times \text{moles}$

$$= 63.4 \times 0.025$$

$$= 1.5875 \text{ g}$$

$$=1.59\text{g}$$

$$\% \text{ Cu} = \frac{1.5875}{2.00} \times 100 = 79.4 \%$$

$$(b)(i) \quad \Delta H_{\text{hyd}} = \Delta H_{\text{stn}} - \text{LE}$$

$$(ii) \quad \Delta H_{\text{stn}} = -(-2327) + (-1920) \\ = +402 \text{ kJ mol}^{-1}$$

(c) - plastics are chemically inert hence cannot react with the acid.

(d) - formation of H_2CO_3 is accompanied by less moles

- therefore high pressure favours the forward reaction

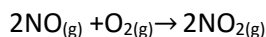
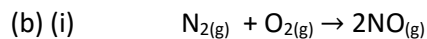
(ii) - backward reaction is favoured; CO_2 is released under high pressure.

4(a) (i) - strong $\text{N}\equiv\text{N}$ bond; high activation energy to break the bond.

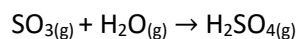
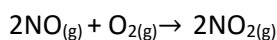
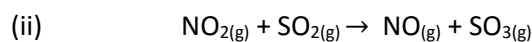
(ii) - lightning release large amounts of heat energy

- Provide enough activation energy to break $\text{N}\equiv\text{N}$ bond;

- O_2 relatively abundant in atmosphere readily reacts.



- High temperatures in engines supply enough energy to break $\text{N}\equiv\text{N}$;



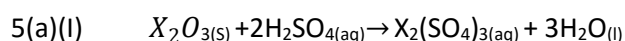
(c) - it is hard;

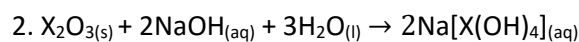
- due to strong covalent bonds.

(ii) - no delocalised electrons

- Hence does not conduct electricity

- Graphite has delocalised electrons along layers.





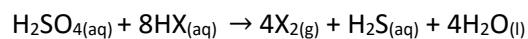
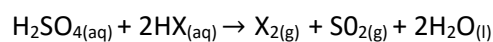
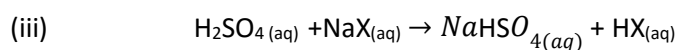
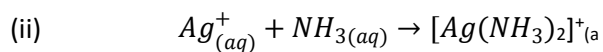
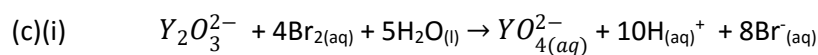
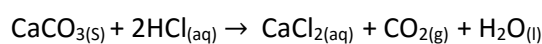
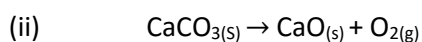
(ii) – hydrolyses vigorously with water;

- Forms an acid solution with $pH = 3$

(b)(i) –thermal stability increases down the group;

- cationic radii increases; leading to a decrease in charge density;

- therefore polarization power decreases; leading to poor distortion of the anion



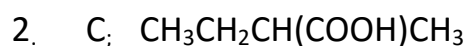
6(a)(i) 2-bromobutane

(ii) 1. Reaction I; reagents- KCN

Conditions- reflux in ethanol

2. Reaction II; reactants- $\text{LiAl}(\text{H})_4$ in ether

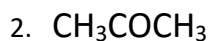
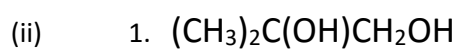
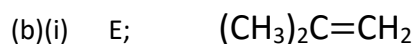
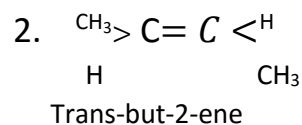
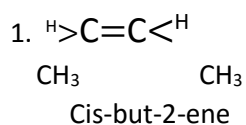
(iii)



(b)(i) reagents; KOH

Conditions; solvent of ethanol

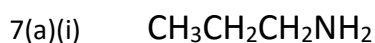
1.
H



(iii) 1. Purple colour of KMnO_4 is decolourised

(iv) 2.- Purple colour decolourised

- bubbles of CO_2 gas evolved



(ii) - reacts with acids in a neutralization reaction.

(iii) - Propylamine

- pH is alkaline;

- Nitrogen has a lone pair of electrons;

- Capable of accepting H^+ ions;

2-aminopropanoic acid

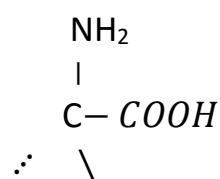
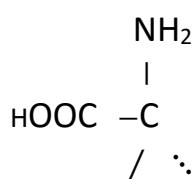
- Amino acid, undergoes self neutralisation;

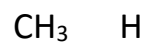
- Solution becomes neutral

(b)(i) - contains central carbon atom bonded to four different groups.

(ii) - optical isomerism.

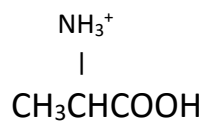
(iii)





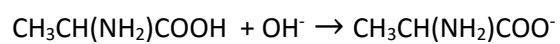
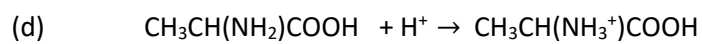
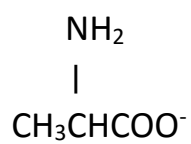
(c)(i)-species with both positive and negative charges of equal size.

(ii) 1.

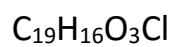


2.

2.



8(a)(i)



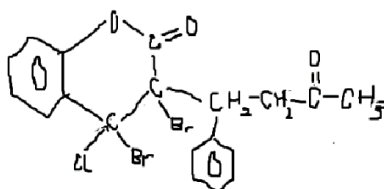
(ii) - alkene

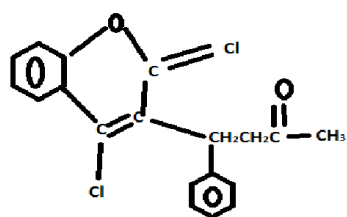
-Benzene ring

-Ketone

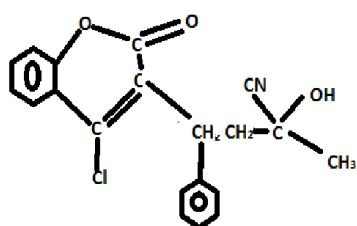
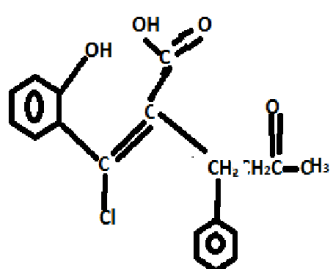
-Ester

(b)(ii) 1.





2.



N

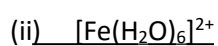
- (ii) 1. – electrophilic substitution
2. – nucleophilic substitution
3. –hydrolysis
4. –nucleophilic addition

(c)



9(a)(i) bidentate- species which form two dative bonds with a transition metal element.

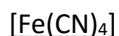
Polydentate- species which form two or more dative bonds with transition elements.



- Single lone electrons in valence shells; low spin electrons/ weak magnetic field

Colour

- Fe^{2+} has empty orbitals; capable of accommodating a high energy electron;
- Promotion of electron leads to absorption of energy and complementary colour is displayed.

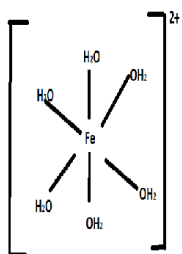


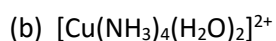
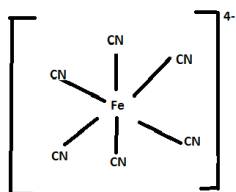
- CN^- ligand pushes electrons in Fe^{2+} , forcing them to be paired;
- Paired electrons cause diamagnetism.

Colour

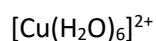
- Electrons are paired due to strong CN^- ligands;
- Very large energy difference between low and high energy d-orbitals.

(iii)





- Has a deep blue colour.
- promotion of high energy electrons to high energy d-orbitals.



- is paramagnetic.
- Has singly paired electrons

(i)

Thin layer	Gas liquid
-separation of non-volatile substances	-separation of gases and volatile liquids
-mobile phase is a liquid	-mobile phase is a gas.
-results appear on a chromatogram	-results are obtained by a detector

(ii) - analysis of beverages

- Analysis of petroleum fuels

(d) - can be easily delivered to the target organ directly.

- they are very small therefore can reach deeper in cells.

10(a)(i) – electrophoresis

(ii) 1. R- no net charge therefore remains stationary.

2.- S has a big positive charge; Q has a small negative charge

- S travels a large distance towards the negative electrode;
- S therefore has more basic groups;
- Q travels a short distance towards the positive end;

- Q hence possess a small number of acidic groups.

(iii)-the movement of amino acids is affected by pH

-each amino acid has its own isoelectronic point.

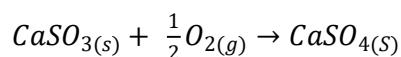
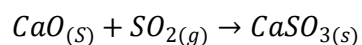
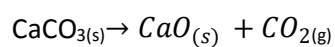
(iv)-spray the buffered paper with ninhydrin;

-Patterns of separation are revealed

(b)(i)-beds of crushed limestone are prepared;

- limestone decomposes to CaO due to high temperatures;

- CaO neutralizes acidic SO₂ gas;



(ii)-leads to formation of acid rain; destroys leafy crops like tobacco;

- Erodes structures made of limestone;

-Is a respiratory irritant.

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